

THE PHYSICAL AI CURRICULUM BLUEPRINT & ROADMAP

Systematic Deconstruction: From Causal Foundations to Pipeline Organs & Real-World Assurance

PART 1: THE FOUNDATIONAL TRIAD (THE REALM OF PHYSICS & LATENCY)

CHAPTER 1

Physical Causality

Causal boundary, 3 criteria,
irreversible mutation ($W_t \rightarrow W_{t+1}$)

CHAPTER 2

Time & Latency Metrology

7-stage latency ledger, tail P99,
freshness wall & stopping distance

CHAPTER 3

The Agent Workflow

Great Tug-of-War, 9-station lifecycle,
3 cadences & multi-rate contracts

PART 2: THE 7 CANONICAL PIPELINE ORGANS (BUILDING THE WORKFLOW)

CHAPTER 4

Perception

MIPI DMA stream
ViT / DINOv2 tokens
3D SE(3) affordances

[Stations 2 & 3]

CHAPTER 5

World Models

Latent JEPAs
Coordinate trees
Uncertainty bounds

[Station 4]

CHAPTER 6

Semantic Intent

1 Hz VLMs
Open-world goals
Expiring leases (TTL)

[Stations 1 & 5]

CHAPTER 7

Action Chunking

Diffusion / ACT
Delay amortization
 C^2 jerk continuous

[Station 6]

CHAPTER 8

Real-Time Reflex

1 kHz CBF safety QP
Proposal-permission
IEC dynamic halts

[Station 7]

PART 3: INTEGRATION, GOVERNANCE & DEPLOYMENT (SYSTEM QUALIFICATION)

CHAPTER 9

Workload Placement

MPU vs MCU vs NPU
UMA contention & IPC

CHAPTER 10

Human Governance

Bumpless manual takeover
Intervention tagging

CHAPTER 11

Assurance & Release

STPA hazard mitigation
Claim-Argument-Evidence

CHAPTER 12

Capstone Deployment

Full dual-brain release
Arduino UNO Q bench run